Hybrid Quantum–Classical Physics-Informed Neural Networks for Accelerating Multiphysics Simulation

Fourier features, data re-uploading and parameterised quantum circuits

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Outline

1. Motivation
2. Background
3. Proposed framework
4. Experimental setup
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Accurate simulation is too slow for design loops

- High-fidelity multiphysics CFD takes hours to days per run.
- The real cost is the parameter sweep: every operating point is a new simulation.
- Real-time control and digital twins need full fields in milliseconds.
- Goal: a surrogate that keeps the governing physics, generalises across operating conditions, and runs fast.

Silicon Czochralski (CZ) growth

Melt convection, crucible and crystal rotation, and thermal structure largely set defect density and crystal quality.

LH_2 cryogenic boil-off

Turbulent convection with phase change in cryogenic storage. Planned benchmark; no results yet.

Physics-informed neural networks (PINNs)

- A network $\hat{u}_\theta(x)$ is trained so that the governing PDEs and the boundary and initial conditions are satisfied.
- Derivatives come from automatic differentiation, so no mesh is needed.
- Loss: $\mathcal{L} = \mathcal{L}_{\mathrm{PDE}} + \mathcal{L}_{\mathrm{BC}} + \mathcal{L}_{\mathrm{IC}} + \dots$
- Needs less labelled data and fits coupled multiphysics naturally.
- Limitation: one PINN solves one parameter setting. Changing parameters means retraining, which motivates neural operators (PINO).

Bottleneck 1: spectral bias

- Standard MLPs fit low-frequency structure first. High-frequency content is learned slowly.
- Coupled transport fields have sharp gradients: thermal boundary layers, wall shear, and swirl near rotating walls.
- Classical remedy: Fourier feature mapping of the inputs before the network (Tancik et al., 2020).
- The quantum layer adds a second, trainable nonlinear feature map on top of the Fourier encoding.

Hypothesis to test

The quantum feature map improves high-frequency fit beyond Fourier features alone. This is tested by the ablation planned on the outlook slides.

Bottleneck 2: loss-term imbalance

- Gradient pathologies: a loss term with larger raw gradients dominates training (Wang et al., 2021).
- A low *total* loss can hide poor boundary satisfaction and weak physical consistency.

Trainer (heat equation benchmark)	PDE loss	BC/IC loss	Rel. L_2 error
Fixed weights (baseline)	3.96×10^{-3}	1.15×10^{-3}	0.084
Normalised gradients	9.74×10^{-3}	9.05×10^{-4}	0.054
Adaptive weights	8.22×10^{-7}	2.00×10^{-1}	1.368

Lesson

The adaptive model has the smallest PDE residual and the worst solution. Report the components and an external error, not only the weighted total.

Bottleneck 3: scale of coupled fields

- Five coupled outputs with very different magnitudes: u_r, u_z, u_θ, p, T.
- In the CZ model the pressure coefficient is $p_{\mathrm{coeff}} = p_{\mathrm{scale}}/v_{\mathrm{scale}}^2 = 134{,}128$.
- Velocity-gradient terms scale as $1/Re \approx 0.067$; pressure terms carry p_{coeff}.
- Ratio of pressure to velocity terms is about $2 \times 10^6 : 1$.
- Consequence: the pressure residual can override the data signal.

Why this matters

This mismatch is the main stability problem found in our experiments (see Results).

Hybrid QPINN architecture

Input (r, z)

$\downarrow$

Fourier features $[2 \to 48]$

$\downarrow$

Coordinate projector $[50 \to 128 \to 128]$

$\downarrow$

ResidualMLP $\times 2$ $[128 \to 128]$

Quantum branch

to_quantum $[128 \to 64 \to 16]$, $\pi \tanh(\cdot)$

$\downarrow$

16-qubit circuit, 2 layers

$\downarrow$

LayerNorm(16)

Classical skip

$[128 \to 64]$, tanh

Concatenate $[64 + 16 + 2 = 82] \to$ post network $[82 \to 128 \to 64 \to 5]$
$\to$ hard constraints on u_r and u_θ

Ingredient 1: Fourier feature mapping

- Fixed random projection $B \in \mathbb{R}^{2\times 24}$, entries drawn from $\mathcal{N}(0, 6^2)$ and frozen at initialisation.
- Feature map: $\gamma(x) = [\sin(2\pi xB),\ \cos(2\pi xB)]$, giving 48 features from (r, z).
- Frequencies reach about 15 cycles per unit length in normalised coordinates.
- Frozen, so it adds no trainable weights beyond the 48 entries of B.
- Purpose: give the network access to high-frequency content from the first layer.

Ingredient 2: shallow parameterised quantum circuit

- 16 qubits and 2 variational layers. Inputs enter as angles $\pi \tanh(\cdot) \in [-\pi, \pi]$.

Each layer applies:

1. Data encoding: $R_Y(\theta_i)$ on every qubit.
2. Data re-upload: $R_Z(0.5\,\theta_{(i+\ell) \bmod 16})$.
3. Trainable rotation: $\mathrm{Rot}(\varphi_i, \theta_i, \omega_i)$, 48 parameters per layer.
4. Entanglement: CZ gates in a ring, closing with CZ between qubits 15 and 0.

Readout: $\langle Z_i \rangle$ for $i = 0, \ldots, 15$, giving 16 features, followed by LayerNorm. Total trainable quantum parameters: **96**.

Ingredient 3: data re-uploading

- A variational circuit that encodes its data only once is a low-complexity function of the input.
- Data re-uploading encodes the same inputs again in each layer (Pérez-Salinas et al., 2020).
- The re-upload angle is shifted across qubits, $\theta_{(i+\ell) \bmod 16}$, so each layer sees a different combination of input features.
- Intended effect: a richer input–output function from a shallow circuit.
- On hardware the circuit has about 50 gates per shot.

Design rule

The quantum layer is a feature-extraction module. The classical network carries most of the computation.

Parameter budget: the quantum layer is deliberately small

Component	Parameters
Coordinate projector (2 layers)	23,040
ResidualMLP blocks (2×2 layers)	66,048
to_quantum (2 layers)	9,296
Classical skip	8,256
Quantum circuit weights	**96**
Quantum LayerNorm	32
Post network (3 layers)	19,205
Fourier matrix B (frozen)	48
Total	**126,021**
Quantum fraction	**0.076%**

Implication

Any performance difference must be traced to the quantum layer by a controlled ablation, not assumed.

Governing equations (1/2): mass and momentum

Steady, axisymmetric, incompressible Navier–Stokes, nondimensional.

Continuity:

$$\frac{\partial (r\, u_r)}{\partial r} + r\, \frac{\partial u_z}{\partial z} = 0$$

Radial momentum:

$$u_r u_{r,r} + u_z u_{r,z} - \frac{u_\theta^2}{r} + p_{\mathrm{coeff}}\, p_{,r} - \frac{1}{Re}\left[u_{r,rr} + \frac{u_{r,r}}{r} - \frac{u_r}{r^2} + u_{r,zz}\right] = 0$$

Axial momentum (with buoyancy):

$$u_r u_{z,r} + u_z u_{z,z} + p_{\mathrm{coeff}}\, p_{,z} - \frac{1}{Re}\left[u_{z,rr} + \frac{u_{z,r}}{r} + u_{z,zz}\right] - \frac{Gr}{Re^2}\,\theta = 0$$

Governing equations (2/2): swirl and energy

Swirl (angular momentum):

$$u_r u_{\theta,r} + u_z u_{\theta,z} + \frac{u_r u_\theta}{r} - \frac{1}{Re}\left[u_{\theta,rr} + \frac{u_{\theta,r}}{r} - \frac{u_\theta}{r^2} + u_{\theta,zz}\right] = 0$$

Energy:

$$u_r \theta_r + u_z \theta_z - \frac{1}{Pr\ Re}\left[\theta_{rr} + \frac{\theta_r}{r} + \theta_{zz}\right] = 0$$

$$\theta = (T - T_{\min})/(T_{\max} - T_{\min}), \quad Re = 15, \quad Pr = 28.463, \quad Gr = 8000, \quad p_{\text{coeff}} = 134{,}128$$

Physics loss: implementation and constraints

- All derivatives via `torch.autograd.grad` with `create_graph=True`.
- Five residual terms (continuity, radial, axial, swirl, energy), normalised by EMA weighting ($\beta = 0.95$).
- Physics weight ramped from 0 at epoch 50 to 0.05 over 200 epochs. Gradient clipping at L2 norm 1.0.
- Hard constraints: $u_r = r \cdot \mathrm{raw}_0$ and $u_\theta = r \cdot \mathrm{raw}_2$, so the velocity vanishes on the axis by construction.

Quantum detachment

The quantum output is detached during the physics loss, so physics gradients do not flow through the quantum circuit. The physics loss therefore cannot directly shape the quantum layer in the current implementation.

Benchmark data: silicon CZ melt

Property	Value
CFD source	Axisymmetric melt solver
Domain	$r, z \in [0, 0.3]$ m
Total points	18,108
Inputs / outputs	(r, z) / u_r, u_z, u_θ, p, T
Temperature range	1685–1750 K
Split (random, spatial)	17,203 train / 905 val

Dimensionless setup

Grashof $Gr = 8000$, Prandtl $Pr = 28.463$, Reynolds $Re = 15$.

Caveat

The reference fields are steady, laminar and axisymmetric. They are controlled reference cases, not fully resolved industrial melt flow.

Evaluation protocol

- **Held-out operating condition:** withhold one whole process setting, not random points.
- **Components:** report unweighted PDE and BC/IC losses separately.
- **Fields:** report error per field, not one pooled score.
- **Matched budgets:** same optimiser, sampling, seed and training budget for every comparison.

Parameter sweep	Range	Held out
T_{hot} (hot wall)	1730–1785 K	1780 K
ω_c (crucible)	-10 to -1 rpm	-4 rpm
ω_s (crystal)	4–20 rpm	7 rpm

Status

The QPINN runs so far used a random spatial split. Generalisation to an unseen operating condition is **not yet tested** for the quantum model.

Training configuration (classical pretraining on GPU)

Hyperparameter	Value
Epochs	400 (about 3.7 h on one H100)
Batch size	256
Learning rate	2×10^{-3}, cosine to 10^{-6}
Physics warm-up	Epoch 50
Physics ramp	200 epochs, $0 \rightarrow 0.05$
Gradient clip	L2 norm 1.0
EMA β	0.95
Checkpoints	Every 50 epochs and best validation

Development path

Built first on a Mac M4 with 8 qubits and one layer. After 120 epochs the validation MSE was about 4.6×10^{-3}. This confirmed the pipeline before GPU training.

Simulation note

The 16-qubit circuit is simulated exactly as a 2^{16}-amplitude statevector on one GPU. That is small enough for classical hardware to handle.

Phase 1: data-only training converges quickly

Epoch	Train loss	Val MSE (total)	Val p	Val T
1	1.16×10^{-2}	4.83×10^{-3}	1.68×10^{-3}	1.37×10^{-3}
10	5.92×10^{-5}	3.11×10^{-4}	7.24×10^{-5}	1.47×10^{-4}
20	5.26×10^{-5}	1.58×10^{-4}	4.62×10^{-5}	5.07×10^{-5}
30	2.96×10^{-5}	1.75×10^{-4}	4.58×10^{-5}	6.15×10^{-5}
50	2.07×10^{-5}	2.13×10^{-4}	1.07×10^{-4}	2.91×10^{-5}

Best checkpoint

Validation MSE 1.58×10^{-4} at epoch 20, in-distribution (random spatial split).

Phase 2: the physics ramp degrades validation error

Epoch	Physics weight	Val p	Val T
50	0 (ramp starts)	$\sim 5 \times 10^{-5}$	$\sim 5 \times 10^{-5}$
60	3.33×10^{-3}	1.05×10^{-1}	7.53×10^{-2}
80	1.00×10^{-2}	1.06×10^{-1}	7.70×10^{-2}
110	2.00×10^{-2}	2.22×10^{-1}	1.14×10^{-1}
200	5.00×10^{-2}	1.06×10^{-1}	1.02×10^{-1}

Critical observation

Once the physics loss activated, validation error for p and T rose from about 5×10^{-5} to about 0.1: roughly three orders of magnitude worse.

Degradation by field and final result

Field	Best MSE (epoch 20)	Final MSE (epoch 400)	Factor
u_r	1.93×10^{-6}	1.79×10^{-4}	93×
u_z	5.78×10^{-5}	8.83×10^{-3}	153×
u_θ	1.44×10^{-6}	5.06×10^{-5}	35×
p	4.62×10^{-5}	0.100	2,165×
T	5.07×10^{-5}	0.129	2,544×

Final validation MSE is 0.238 at epoch 400, about 1,500× worse than the best checkpoint. Pressure and temperature degrade most.

Root cause: physics-loss instability

```
[0280/400] loss=1.0368e+09 data=5.0592e-02 phys=2.1591e+10
[0290/400] loss=1.1795e+14 data=4.9662e-02 phys=2.1481e+15
[0300/400] loss=5.5708e-02 data=4.8543e-02 phys=9.6150e-02
```

- Data loss stayed near 0.05. The explosion sits entirely in the physics residual.
- Gradient clipping limited the damage; the model recovered by epoch 300.
- Cause: the p_{coeff} scale mismatch from Bottleneck 3.

Candidate fixes (to be tested)

Physics weight 0.001–0.005; per-field weighting of the pressure residual (candidate $1/p_{\mathrm{coeff}}^2 \approx 5.6 \times 10^{-11}$); re-check the pressure nondimensionalisation.

Stage 3: why fine-tune on IBM quantum hardware?

Hardware noise

- Gate fidelity on ibm_fez: about 99.5% single-qubit and 99% two-qubit.
- Readout error: about 1–2% per qubit.

Consequence

Hardware $\langle Z_i \rangle$ differs systematically from simulated values. The pretrained head receives shifted inputs and degrades.

Adaptation

Freeze the quantum circuit. Retrain only the classical post-network head on hardware outputs.

Parameter	Value
Trainable scope	Head only (19,205)
Calibration points	8 (near-axis and near-wall)
Batch size	1 sample per job
Optimiser	Adam, lr 10^{-3}
Shots per circuit	4,096
Planned epochs	100

IBM hardware execution: 500 jobs on ibm_fez

Backend property	Value
Processor	156-qubit Eagle r3
Error mitigation	M3 twirling
Queue at submission	0 pending jobs
Instance	hybrid_qcpinn (flex)

Execution metric	Value
Jobs completed / failed	500 / 0
Total shots	2,048,000
Session duration	125.0 min
Throughput	4.0 jobs/min
Forward passes equivalent	about 62

Session ended by plan limit

The session stopped at 01:23 UTC+2 when the flex-plan time ran out, at about 62 of 100 planned epochs. All 500 results are stored on IBM servers under saved job IDs. A second instance was also exhausted.

Hardware output sanity check

Qubit	$\langle Z \rangle$	σ reported	σ shot-noise estimate
Q0	$+0.3417$	0.0122	0.0147
Q1	$+0.7645$	0.0136	0.0101
Q5	-0.8555	0.0088	0.0081
Q13	-0.9077	0.0070	0.0066

Shot-noise estimate: $\sigma_i = \sqrt{(1 - \langle Z_i \rangle^2)/4096}$. The reported spread is close to this estimate, with deviations up to about 35%. Hardware drift and gate noise are not yet separated from sampling noise.

Trained quantum parameters and classical weights

Rotation angles (best checkpoint)

- Mean 3.49 rad, std 1.79 rad, range 0.003 to 6.27 rad.
- Spread across the full $[0, 2\pi]$ range.
- No qubit stuck at a degenerate value (0 or π).

Quantum LayerNorm

Scale $\gamma \in [0.94, 1.00]$, bias within ± 0.03. At epoch 20 the qubits have not yet been differentiated, as expected for the data-only phase.

Classical layers

All well conditioned: zero mean, standard deviation 0.17 to 0.36 for hidden layers.

Reading

The circuit uses its full rotational range, but the quantum layer's contribution to accuracy is still unmeasured.

Claims we can make today

Claim	Status	Evidence
Pipeline runs end to end on GPU and real QPU	**Shown**	500/500 IBM jobs completed, none failed
Data-only model fits CZ fields in distribution	**Shown**	Val MSE 1.58×10^{-4}, random split
Physics loss improves field accuracy	Not shown	Validation error rose 10^2–$10^3\times$ at weight 0.05
Quantum layer adds accuracy over classical	Not shown	Ablation not yet run
Faster convergence than classical PINN/PINO	Not shown	No matched classical baseline yet
Better generalisation to new conditions	Not tested	No case-wise holdout for QPINN
Lower computational cost	Not shown	H100 3.7 h, QPU 2.1 h; no classical timing
Quantum computational advantage	Not claimed	16-qubit statevector is classically simulable
Concentration and evolving interfaces	Out of scope	Single-phase, steady, laminar, prescribed surface
LH_2 boil-off results	Planned	Benchmark defined, no data yet

Critical next experiments

Priority	Experiment	Compute
Critical	Classical-only baseline with matched parameter budget	$1\times$ H100, about 4 h
Critical	Random-feature control in place of the quantum circuit	$1\times$ H100, about 4 h
Critical	Retrain with physics weight 0.001–0.005	$1\times$ H100, about 4 h
Critical	Retrieve the 500 IBM job results (API)	Minimal
High	Case-wise holdout on the three CZ sweeps	3 training runs
High	Per-field physics weighting (pressure residual)	Architecture change
High	Complete head fine-tuning on IBM, renewed plan	About 500 more jobs
Medium	LH_2 boil-off benchmark with CFD reference data	CFD + H100

Path to real-time digital twins

Now	Next	Then	Goal
End-to-end pipeline validated on GPU and IBM hardware. Best data-only checkpoint in hand.	Physics-consistent training: weight fix, per-field weights, and ablation against classical baselines.	Generalisation: case-wise holdout, parametric inputs (Re, rotation rates, pull rate), and the LH_2 benchmark.	Digital twin: agentic CFD generates verified training cases from one prompt, a physics validator checks the outputs, and CPU inference serves the surrogate.

Key takeaways

1. The hybrid QPINN trains end to end; the quantum layer is 0.076% of all parameters.
2. The data-only model reaches validation MSE 1.58×10^{-4}. The physics term at weight 0.05 degraded pressure and temperature, a scaling problem to fix.
3. The full pipeline ran on IBM `ibm_fez`: 500 jobs, 2.05 million shots, no failures.
4. Convergence, generalisation and cost advantages over classical PINN/PINO are hypotheses until the controlled ablations are done.
5. Report loss components and per-field errors. Aggregate scores hide failures.

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Thank you

Questions?